Longbin Zhang, Ph.D.

Professor, School of Artificial Intelligence and Robotics, Hunan University

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Employment History

2025.03 – Present — **Professor**, School of Artificial Intelligence and Robotics, Hunan University, China.

2023.03 – 2025.02 — **Research Fellow**, Rehabilitation Research Institute of Singapore, School of Mechanical and Aerospace Engineering, Nanyang Technological University, Singapore.

2021.09 – 2023.02 — **Postdoctoral Researcher**, KTH MoveAbility Lab, Department of Engineering Mechanics, KTH Royal Institute of Technology, Stockholm, Sweden.

Education

2017.09 – 2021.10 — **Ph.D. Engineering Mechanics (Biomechanics)**, KTH Royal Institute of Tech.
Thesis: *Biomechanical parameter estimation for wearable exoskeleton system design.*

2014.09 – 2017.06 — **M.S. Control Engineering**, South China University of Technology.
Thesis: *Controller design for flexible robotics under restrictions.*

2010.09 – 2014.06 — **B.S. Automation**, Southwest University of Science and Technology.

Research Interests

Human–Robot Interaction, Wearable and Rehabilitation Robotics, Human Movement Analysis, Neuromusculoskeletal Modeling, Artificial Intelligence and Machine Learning, Multimodal Sensor Fusion, Modeling and Simulation, Movement Intention Recognition, Personalized Assistive Control.

Honors and Awards

2026 — **Best Paper Presentation Award**, IEEE International Conference on Advanced Robotics and Mechatronics (ICARM), Paris, France

2024 — **Travel Scholarship**, Scientific Meetings of the Australian and New Zealand Society of Biomechanics and the Australian and New Zealand Orthopaedic Research Society, Australia

2022 — **Travel Scholarship**, European Society for Movement Analysis in Adults and Children, Ireland

2021 — **Best Conference Paper Finalist**, IEEE International Conference on Advanced Robotics and Mechatronics (ICARM), Chongqing, China

2020 — **Best Conference Paper Finalist (Co-author)**, IEEE International Conference on Advanced Robotics and Mechatronics (ICARM), Shenzhen, China

2019 — **Travel Scholarship**, XXVII Congress of International Society of Biomechanics, Canada

Travel Scholarship, IRCN Neuro-inspired Computation Course, Japan

Travel Scholarship, Winter School on Wearable Robotics (EU COST Action network), Slovenia

2014 — Sichuan Province **Outstanding Graduate**, Sichuan, China

Research Projects

2027.01–2030.12 | Human-in-the-Loop Adaptive Control of Orthopedic Rehabilitation Robots Integrating Biomechanical Sensing

Principal Investigator (PI), National Natural Science Foundation of China (NSFC), General Program, Grant No. 62673192. Funding: RMB 500,000.

2026.05–2029.04 | Development of an Integrated Stroke Rehabilitation System Based on an EEG Decoding Foundation Model
Principal Investigator (PI), Key Research and Development Program of Hunan Province, Grant No. 2026QK3021. Funding: RMB 2,700,000.

2026.01–2028.12 | Mechanism- and Data-Driven Intelligent Wearable Rehabilitation System for Patients with Knee Osteoarthritis
Principal Investigator (PI), National Natural Science Foundation of China (NSFC), Young Scientists Fund (Category C), Grant No. 62503165. Funding: RMB 300,000.

2025.08–Present | AI-Based Intelligent Orthopedic Rehabilitation Robotic System Development and Clinical Translation
Co-Principal Investigator (Co-PI), Changsha Municipal Health Commission, Regional Medical–Academic–Industry Collaborative Research Program. Total project funding: RMB 2,300,000.

2025.01–2027.12 | Biomechanics-Based Intelligent Wearable Technologies
Principal Investigator (PI), National Natural Science Foundation of China (NSFC), Excellent Young Scientists Fund (Overseas). Funding: RMB 2,000,000.

2022.06–2023.06 | Neuromusculoskeletal Model-Informed Artificial Neural Networks for Real-Time Joint Torque Prediction
Principal Investigator (PI), Promobilia Foundation, Sweden, Grant No. A22078. Funding: SEK 372,200.

2021.06–2025.02 | Big Data-Driven Research on Human Mobility and Balance
Project Participant, Rehabilitation Research Institute of Singapore (RRIS), Singapore.

2020.06–2021.06 | Design and Control of a Real-Time EMG-Driven Wearable Knee Assistive Device
Principal Investigator (PI), Biomechanics Modeling and Experimentation Center, KTH Royal Institute of Technology, Sweden, Project No. 770024. Funding: SEK 55,000.

2019.01–2024.01 | BADASS: BiomechAnics in Motion Disorders and ASSistance
Project Participant, Swedish Research Council, Sweden, Grant No. 2018-00750. Total project funding: SEK 12,000,000.

2019.01–2023.01 | Multiscale and Multimodal Biomechanical Research for Persons with Motion Disorders
Project Participant, Promobilia Foundation, Sweden, Grant No. 18200. Total project funding: SEK 11,500,000.

2018.12–2019.12 | Quantitative Assessment of Muscle Architecture and Stiffness Following Botulinum Toxin Injections in Individuals with Post-Stroke Hemiparesis
Project Participant, Promobilia Foundation, Sweden, Reference No. 18014. Total project funding: SEK 350,000.

2018.01–2022.01 | INC OME: An Integrated Neuromusculoskeletal Modeling Framework
Project Participant, Swedish Research Council, Sweden, Grant No. 2018-04902. Total project funding: SEK 3,600,000.

2016.07–2019.06 | Development of a Wearable Bionic Leg with Elastic Actuators
Project Participant, Ministry of Science and Technology of China, Project No. 2016A020220003. Total project funding: RMB 500,000.

Research Publications

Journal Articles

1. **L. Zhang**, A. Sidarta, T.-L. Wu, P. Jatesiktat, H. Wang, L. Li, P. W.-H. Kwong, A. Long, X. Long, and W. T. Ang, "Towards clinical application of enhanced timed up and go with markerless motion capture and machine learning for balance and gait assessment," *IEEE Journal of Biomedical and Health Informatics*, vol. 30, no. 8, pp. 7072–7081, 2026.

2 Z. Li, Z. Wan, X. Fu, S. Wang, T. Zhang, H. Zhang, **L. Zhang**, X. Zhu, and R. Wang, "Physics-informed multimodal learning for wearable estimation of ground reaction forces and lower-limb joint moments," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 2026, Submitted; corresponding authors: L. Zhang, X. Zhu, and R. Wang.

3 R. Wang, **L. Zhang**, H. Jalo, O. Tarassova, G. V. Pennati, and A. Arndt, "Individualized muscle architecture and contractile properties of ankle plantarflexors and dorsiflexors in post-stroke individuals," *Frontiers in Bioengineering and Biotechnology*, vol. 12, pp. 1–12, 2024.

4 **L. Zhang**, T. V. Wouwe, S. Yan, and R. Wang, "Emg-constrained and ultrasound-informed muscle-tendon parameter estimation in post-stroke hemiparesis," *IEEE Transactions on Biomedical Engineering*, pp. 1–12, 2024.

5 **L. Zhang**, D. Soselia, R. Wang, and E. M. Gutierrez-Farewik, "Estimation of joint torque by emg-driven neuromusculoskeletal models and lstm networks," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 31, pp. 3722–3731, 2023.

6 **L. Zhang**, X. Zhang, X. Zhu, R. Wang, and E. M. Gutierrez-Farewik, "Neuromusculoskeletal model-informed machine learning-based control of a knee exoskeleton with uncertainties quantification," *Frontiers in Neuroscience*, vol. 17, pp. 1–10, 2023.

7 **L. Zhang**, X. Zhu, E. M. Gutierrez-Farewik, and R. Wang, "Ankle joint torque prediction using an nms solver informed-ann model and transfer learning," *IEEE Journal of Biomedical and Health Informatics*, vol. 26, no. 12, pp. 5895–5906, 2022.

8 **L. Zhang**, D. Soselia, R. Wang, and E. M. Gutierrez-Farewik, "Lower-limb joint torque prediction using lstm neural networks and transfer learning," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 30, pp. 600–609, 2022.

9 **L. Zhang**, Y. Liu, R. Wang, C. Smith, and E. M. Gutierrez-Farewik, "Modeling and simulation of a human knee exoskeleton's assistive strategies and interaction," *Frontiers in Neurorobotics*, p. 13, 2021.

10 **L. Zhang**, Z. Li, Y. Hu, C. Smith, E. M. Gutierrez-Farewik, and R. Wang, "Ankle joint torque estimation using an emg-driven neuromusculoskeletal model and an artificial neural network model," *IEEE Transactions on Automation Science and Engineering*, vol. 18, no. 2, pp. 564–573, 2021.

11 **L. Zhang**, W. Qi, Y. Hu, and Y. Chen, "Disturbance-observer-based fuzzy control for a robot manipulator using an emg-driven neuromusculoskeletal model," *Complexity*, vol. 2020, pp. 1–10, 2020.

12 **L. Zhang**, Z. Li, and C. Yang, "Adaptive neural network based variable stiffness control of uncertain robotic systems using disturbance observer," *IEEE Transactions on Industrial Electronics*, vol. 64, no. 3, pp. 2236–2245, 2017.

13 M. Romanato, **L. Zhang**, Z. Sawacha, and E. M. Gutierrez-Farewik, "Influence of different calibration methods on surface electromyography-informed musculoskeletal models with few input signals," *Clinical Biomechanics*, 2023.

14 Y.-X. Liu, **L. Zhang**, R. Wang, C. Smith, and E. M. Gutierrez-Farewik, "Weight distribution of a knee exoskeleton influences muscle activities during movements," *IEEE Access*, vol. 9, pp. 91 614–91 624, 2021.

15 Y. Zhao, S. Yan, **L. Zhang**, B. Shi, and L. Yang, "Do gait patterns normalize concurrently with the recovery of foot arches in children with flatfoot? a prospective cohort study," *Journal of Biomechanics*, vol. 187, pp. 1–9, 2025.

16 S. Yan, Y. Zhao, **L. Zhang**, and L. Yang, "Arch-related alteration in foot loading patterns affected by the increasing extent of body mass index in children: A follow-up study," *Gait & Posture*, vol. 100, pp. 247–253, 2023.

17 W. Qi, X. Liu, **L. Zhang**, L. Wu, W. Zang, and H. Su, "Adaptive sensor fusion labeling framework for hand pose recognition in robot teleoperation," *Assembly Automation*, 2021.

18 Z. Li, K. Zhao, **L. Zhang**, X. Wu, T. Zhang, Q. Li, X. Li, and C.-Y. Su, "Human-in-the-loop control of a wearable lower limb exoskeleton for stable dynamic walking," *IEEE/ASME Transactions on Mechatronics*, vol. 26, no. 5, pp. 2700–2711, 2020.

19 Y. Hu, H. Su, **L. Zhang**, S. Miao, G. Chen, and A. Knoll, "Nonlinear model predictive control for mobile robot using varying-parameter convergent differential neural network," *Robotics*, vol. 8, no. 3, p. 64, 2019.

20 J. Chen, H. Su, J. Sandoval, **L. Zhang**, and S. Zhong, "Human inspired robotic intelligence and structure in demanding environments," *Frontiers in Neurorobotics*, vol. 16, p. 1 026 917, 2022.

21 S. E. Ovur, X. Zhou, W. Qi, **L. Zhang**, Y. Hu, H. Su, G. Ferrigno, and E. De Momi, "A novel autonomous learning framework to enhance semg-based hand gesture recognition using depth information," *Biomedical Signal Processing and Control*, vol. 66, p. 102 444, 2021.

22 J. Li, J. Wang, H. Peng, **L. Zhang**, Y. Hu, and H. Su, "Neural fuzzy approximation enhanced autonomous tracking control of the wheel-legged robot under uncertain physical interaction," *Neurocomputing*, vol. 410, pp. 342–353, 2020.

23 X. Zhou, W. Qi, S. E. Ovur, **L. Zhang**, Y. Hu, H. Su, G. Ferrigno, and E. De Momi, "A novel muscle-computer interface for hand gesture recognition using depth vision," *Journal of Ambient Intelligence and Humanized Computing*, vol. 11, pp. 5569–5580, 2020.

24 S. Qiu, Z. Li, W. He, **L. Zhang**, C. Yang, and C.-Y. Su, "Brain–machine interface and visual compressive sensing-based teleoperation control of an exoskeleton robot," *IEEE Transactions on Fuzzy Systems*, vol. 25, no. 1, pp. 58–69, 2016.

25 Y. Wang, H. Kang, D. Wu, W. Yang, and **L. Zhang**, "Global and local spatio-temporal encoder for 3d human pose estimation," *IEEE Transactions on Multimedia*, pp. 1–11, 2023.

26 J. Li, J. Wang, S. Wang, W. Qi, **L. Zhang**, Y. Hu, and H. Su, "Neural approximation-based model predictive tracking control of non-holonomic wheel-legged robots," *International Journal of Control, Automation and Systems*, vol. 19, no. 1, pp. 372–381, 2021.

27 X. Zhang, J. Li, Z. Hu, W. Qi, **L. Zhang**, Y. Hu, H. Su, G. Ferrigno, and E. D. Momi, "Novel design and lateral stability tracking control of a four-wheeled rollator," *Applied Sciences*, vol. 9, no. 11, p. 2327, 2019.

28 H. Su, W. Qi, Y. Hu, J. Sandoval, **L. Zhang**, Y. Schmirander, G. Chen, A. Aliverti, A. Knoll, G. Ferrigno, et al., "Towards model-free tool dynamic identification and calibration using multi-layer neural network," *Sensors*, vol. 19, no. 17, p. 3636, 2019.

29 D. Chen, X. Zhou, J. Li, J. He, X. Yu, **L. Zhang**, and W. Qi, "A muscle teleoperation system of a robotic rollator based on bilateral shared control," *IEEE Access*, vol. 8, pp. 151 160–151 170, 2020.

30 J. Li, J. Wang, S. Wang, H. Peng, B. Wang, W. Qi, **L. Zhang**, and H. Su, "Parallel structure of six wheel-legged robot trajectory tracking control with heavy payload under uncertain physical interaction," *Assembly Automation*, vol. 40, no. 5, pp. 675–687, 2020.

31 Y. Liu, Y.-X. Huang, X. Zhang, W. Qi, J. Guo, Y. Hu, **L. Zhang**, and H. Su, "Deep c-lstm neural network for epileptic seizure and tumor detection using high-dimension eeg signals," *IEEE Access*, vol. 8, pp. 37 495–37 504, 2020.

32 H. Su, Y. Schmirander, S. E. Valderrama-Hincapie, J. Pinedo, X. Zhou, J. Li, **L. Zhang**, Y. Hu, G. Ferrigno, and E. D. Momi, "Asymmetric bimanual control of dual-arm serial manipulator for robot-assisted minimally invasive surgeries," *Sensors and Materials*, vol. 32, no. 4, p. 1223, 2020.

33 W. Qi, H. Su, K. Fan, Z. Chen, J. Li, X. Zhou, Y. Hu, **L. Zhang**, G. Ferrigno, and E. De Momi, "Multimodal data fusion framework enhanced robot-assisted minimally invasive surgery," *Transactions of the Institute of Measurement and Control*, vol. 44, no. 4, pp. 735–743, 2022.

Conference Proceedings

1. Z. Li, S. Wang, T. Zhang, Y. Kong, H. Zhang, **L. Zhang**, A. Sidarta, Y. C. Lim, C. Er, X. Yan, T. Wu, and W. T. Ang, "EEG dynamics and center of pressure modulation during multi-task balance challenges in older adults," in *2026 41st Youth Academic Annual Conference of Chinese Association of Automation (YAC)*, Corresponding authors: L. Zhang and A. Sidarta, IEEE, 2026.

2. X. Fu, Z. Wan, Z. Li, S. Wang, T. Zhang, H. Zhang, **L. Zhang**, X. Zhu, and R. Wang, "Predicting ground reaction forces by LSTM neural networks with multi-modal data," in *2026 41st Youth Academic Annual Conference of Chinese Association of Automation (YAC)*, Corresponding authors: L. Zhang, X. Zhu, and R. Wang, IEEE, 2026.

3. Y. Xie, W. Yuan, Z. Li, S. Wang, T. Zhang, X. Shen, L. Zhang, T. Wei, P. Jatesiktat, T.-L. Wu, Y. Kong, W. Wei, Y. Furletov, **L. Zhang**, and W. T. Ang, "Automatic FMA-UE upper-limb assessment using monocular RGB video and body-hand keypoint fusion," in *2026 China Automation Congress (CAC)*, Corresponding author: L. Zhang, IEEE, 2026.

4. Z. Liu, J. Zhang, H. Xue, Z. Li, L. Li, S. Wang, L. Zhang, X. Shen, T. Zhang, W. Wei, Y. Furletov, and **L. Zhang**, "Disturbance-observer-based second-order adaptive nonsingular terminal sliding mode control for exoskeletons," in *2026 International Conference on Robotics and Learning (ICRL)*, Corresponding authors: S. Wang, L. Zhang (seventh author), and L. Zhang (last author); author initials require confirmation, 2026.

5. **L. Zhang**, Z. Li, X. Fu, Y. Xie, Z. Liu, X. Yan, S. Wang, T. Zhang, H. Zhang, K. Yang, T.-L. Wu, P. Jatesiktat, A. Sidarta, and W. T. Ang, "Detecting heel strike and toe off events using kinematic methods and LSTM models," in *2026 11th IEEE International Conference on Advanced Robotics and Mechatronics (ICARM)*, Best Presentation Award; corresponding author: Z. Li, IEEE, 2026.

6. **L. Zhang**, X. Zhang, X. Zhu, R. Wang, and E. M. Gutierrez-Farewik, "Knee joint torque prediction with uncertainties by a neuromusculoskeletal solver-informed gaussian process model," in *2023 International Conference on Advanced Robotics and Mechatronics (ICARM)*, IEEE, 2023, pp. 1035–1040.

7. **L. Zhang**, X. Zhu, E. M. G. Farewik, and R. Wang, "Estimation of ankle dynamic joint torque by a neuromusculoskeletal solver-informed nn model," in *2021 6th IEEE International Conference on Advanced Robotics and Mechatronics (ICARM)*, IEEE, 2021, pp. 75–80.

8. **L. Zhang**, Y. Hu, H. Su, J. Li, and S. E. Ovur, "Adaptive neural network control for a lower-limb exoskeleton using variable stiffness transferring," in *2020 5th International Conference on Advanced Robotics and Mechatronics (ICARM)*, IEEE, 2020, pp. 240–245.

9. Y. Hu, G. Chen, **L. Zhang**, H. Su, M. Li, Y. Schmirander, H. Cao, and A. Knoll, "Fuzzy adaptive control-based real-time obstacle avoidance under uncertain perturbations," in *2020 5th International Conference on Advanced Robotics and Mechatronics (ICARM)*, IEEE, 2020, pp. 50–55.

10. **L. Zhang**, Z. Chen, Z. Li, C.-Y. Su, and Z. Xiao, "Adaptive neural network control for uncertain mimo robotic systems with time-varying delay and unknown backlash-like hysteresis," in *2015 IEEE International Conference on Information and Automation*, IEEE, 2015, pp. 1827–1832.

11. C. Shi, **L. Zhang**, and Z. Li, "Disturbance observer-based adaptive neural control for a class of uncertain mimo systems with state and input constraints," in *2016 International Conference on Advanced Robotics and Mechatronics (ICARM)*, IEEE, 2016, pp. 426–431.

12. D. Zhong, **L. Zhang**, and P. Zhang, "Ant colony optimization metaheuristic for shortest path of directed graph passing through specified nodes," in *Computer Science and Artificial Intelligence: Proceedings of the International Conference on Computer Science and Artificial Intelligence (CSAI2016)*, World Scientific, 2018, pp. 585–593.

13. Z. Li, W. He, C. Yang, S. Qiu, **L. Zhang**, and C.-Y. Su, "Teleoperation control of an exoskeleton robot using brain machine interface and visual compressive sensing," in *2016 12th World Congress on Intelligent Control and Automation (WCICA)*, IEEE, 2016, pp. 1550–1555.

14 J. Li, J. Wang, S. Wang, H. Peng, Q. Wen, **L. Zhang**, and M. Lin, "A human-robot skill transfer framework of mobile medical robots for autonomous motion with teaching by demonstration," in *2020 5th International Conference on Advanced Robotics and Mechatronics (ICARM)*, IEEE, 2020, pp. 209–213.

Patents

2026 **L. Zhang**, Y. Xie, and Z. Li, *Method and System for Automated Assessment of Post-Stroke Upper-Limb Function Based on Monocular Vision and Multi-Model Fusion*. Chinese patent application No. 202611010771.8, filed July 8, 2026. (Pending)

X. Wang, **L. Zhang**, Z. Zhang, Y. Jia, and Z. Li, *Multimodal Dental Diagnosis and Treatment Data Management System and Method*. Chinese patent application No. 202611702764.4, filed October 30, 2026. (Pending)

2018 F. Peng, H. Wen, Z. Li, and **L. Zhang**, *A Continuously Adjustable Joint Based on a Floating Spring*. Chinese patent No. ZL 201610799641.7, September 14, 2018.

2017 F. Peng, Z. Li, W. Ye, C. Yang, and **L. Zhang**, *A Variable-Stiffness Robotic Elastic Joint*. Chinese patent No. ZL 201610173210.X, June 20, 2017.

2016 W. Ye, Z. Li, F. Peng, **L. Zhang**, and C. Yang, *A Dual-Motor-Driven Variable-Stiffness Elastic Joint for Robots*. Chinese utility model patent No. ZL 201620232982.1, October 5, 2016.

Academic Activity

2021.06-now **Review Editor**, Frontiers in Neuroscience (IF 4.3)

Review Editor, Frontiers in Neurorobotics (IF 3.1)

2023 **Chief Guest Associate Editor**, special issue on Frontiers in Neuroscience: Advanced Technology for Human Movement Rehabilitation and Enhancement

2021 **Guest Associate Editor**, special issue on Frontiers in Neurorobotics: Human Inspired Robotic Intelligence and Structure in Demanding Environments

Reviewers for around 20 journals, such as IEEE Trans. on Industrial Electronics, IEEE Trans. on Biomedical Engineering, IEEE Trans. on Automation and Engineering, IEEE Trans. on Cybernetics, IEEE Robotics and Automation Letters, IEEE Trans. on Neural systems and Rehabilitation Engineering